

questions 1,2 consider the current through a 1 mH inductor is  $i(t)=60 \cos 100t$  mA.  
terminal voltage across the inductor is:

2) The Energy stored in the inductor is:

**Prob 2:** For questions 3,4,5,6 consider the circuit shown in figure 1. The switch has been closed for a long time. At  $t=0$ , the switch is opened. Answer:

3) The initial current  $i(t=0+)$  is:

4) The time constant of the circuit is :

5) The current in the inductor at  $t= 500$  ms is:

6) The final value of the voltage  $v(t=\infty)$  across the inductor is :

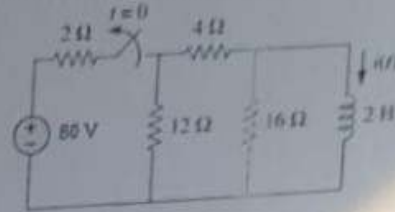


Fig.1

**Prob. 3:** For the questions 7,8,9,10 consider the circuit shown in figure 2 and answer.

7) The value of  $V_0$  is:

8) The value of the current (i) in 5 Ω is:

9) The power absorbed by 3 Ω resistor is:

10) The power absorbed by 12 V voltage source is:

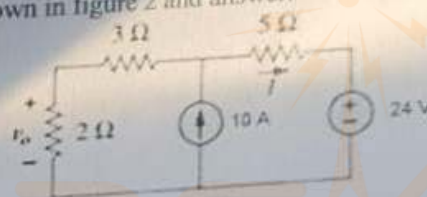


Fig.2

**Prob. 4:** Consider the circuit shown in figure 3. Use Thevenin's theorem with respect to terminals (a - b) and answer:

a. The Thevenin voltage is .....

b. The Thevenin resistance is.....

c. The current I is.....

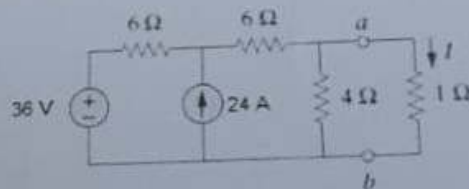


Fig. 3

1. What is the value of the load resistance that should be connected to the ab terminals to achieve maximum power transfer?

a) Refer to figure 4; the value of the equivalent resistance is:

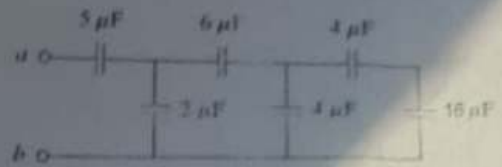


Fig. 4

b) Refer to figure 5; the value of the equivalent resistance is:

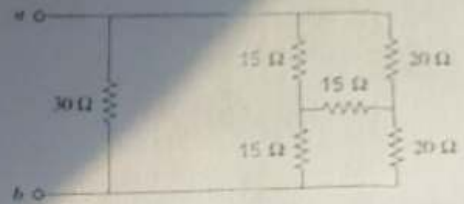


Fig. 5

Good Luck

